

# Talha Jobayer Zihan

Rajshahi, Bangladesh • [jobayertalha2020@gmail.com](mailto:jobayertalha2020@gmail.com) • +8801721577792 • [linkedin.com/in/talha-jobayer](https://www.linkedin.com/in/talha-jobayer) • [github.com/jobayertalha](https://github.com/jobayertalha) • [Portfolio](#)

## SUMMARY

AI/ML engineer, CSE graduate (RUET) with two IEEE-published NLP papers and hands-on experience shipping ML systems end to end: RAG pipelines, transformer fine-tuning, and PyTorch models served through FastAPI and React apps.

## EXPERIENCE

- Research Intern, ELITE Research Lab**Jun 2026 – Present
- R&D in **AI and Machine Learning**: literature review, model development, and experimentation.
- Industrial Training, Netro Systems**Rajshahi | 2 Weeks
- Used industry-standard development tools and collaborated with developers on real projects.
- Industrial Attachment, Frontier Semiconductor BD Ltd**Dhaka | 2 Weeks
- Observed semiconductor fabrication processes and quality control.

## PROJECTS

- CareerVector – RAG-Based CV Analyzer**[GitHub](#)
- Matches CVs to job descriptions with sub-second retrieval; cut query latency by **80%** and lifted top-1 precision by **35%** over a TF-IDF baseline.
  - Stack**: LangChain, FAISS (IVF), OpenAI Embeddings, Python.
- Cross-Domain NIDS – Network Intrusion Detection Platform**[GitHub](#)
- Detects attacks in PCAP/PCAPNG traffic across different network domains using transfer learning (**Lopez17CNN**, **BiC**, **TDNR**), with an analyst dashboard for uploads, threats, and alerts.
  - Stack**: PyTorch, FastAPI, React, TypeScript, Vite; deployed on Vercel and Render.
- Bangla Sarcasm Detection using Transformers**[Kaggle](#)
- Classifies sarcasm in Bangla text at **88% F1** (10k balanced samples); regularization cut overfitting by **15%** and converged in 3 fewer epochs.
  - Stack**: BanglaBERT, Mixup, FGM adversarial training, R-Drop, OneCycleLR.
- ResearchFlow – Research Task Management Platform**[GitHub](#)
- Tracks multiple papers, tasks, conferences, and deadlines in separate workflows, with AI-generated progress insights and next-task suggestions.
  - Stack**: React, Vite, FastAPI, PostgreSQL, Supabase, Groq AI; Vercel and Cloudflare Workers (Hyperdrive).
- Image Classifier using CNN**TensorFlow / Keras
- Classifies handwritten digits at **95% test accuracy** on 60k MNIST samples.
  - Stack**: TensorFlow, Keras, 3-layer CNN with batch normalization.
- Car Rental System Web App**[GitHub](#)
- Manages car bookings with real-time date validation, booking history, secure login, and an admin panel.
  - Stack**: PHP, MySQL, HTML/CSS, JavaScript.

## RESEARCH WORK

- Fast Gradient Method (FGM) Fine-Tuning for Bengali Sarcasm Detection**[IEEE Xplore](#)
- Proposed FGM adversarial fine-tuning on BanglaBERT; presented at ICECTE 2026 (Certificate of Contribution).
- Mental Health Text Classification via Data Augmentation and Transformer Ensemble**[IEEE Xplore](#)
- Improved classification accuracy using data augmentation and transformer model ensembling.

## TECHNICAL SKILLS

**AI / ML**: PyTorch, TensorFlow, Transformers, BanglaBERT, CNNs, RAG, LangChain, FAISS, NumPy, Pandas  
**Programming**: Python, C++, C, Java, SQL, PHP  
**Web / Backend**: FastAPI, React, Next.js, TypeScript, Vite, Tailwind CSS, HTML/CSS, JavaScript  
**Tools**: Git, GitHub, Docker, Linux, Kaggle, MySQL, Arduino, Vercel, Render

## EDUCATION

- Rajshahi University of Engineering and Technology (RUET)**2022 – 2026
- B.Sc. in Computer Science and EngineeringCGPA: 3.64 / 4.00
- New Govt. Degree College, Rajshahi**2018 – 2020
- Higher Secondary Certificate (HSC)GPA: 5.00 / 5.00

## CERTIFICATIONS

[Machine Learning in Production](#) (DeepLearning.AI) • [NLP in TensorFlow](#) (DeepLearning.AI) • [Deep Learning & Reinforcement Learning](#) (IBM) • [Machine Learning with Python](#) (IBM) • [Intro to Neural Networks and PyTorch](#) (IBM) • [Agents Course](#) (Hugging Face) • [PHP \(Laravel\) Training](#) – RUET CSE, EDGE (80 hrs)

## LEADERSHIP & VOLUNTEERING

- Officer – Marketing, TEDxRUET**2024 – 2025
- Co-organized a TEDx event; managed marketing outreach and audience engagement.